

Syed Afraz

Machine Learning Engineer

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Education

NUST Oct 2020 – June 2024
Bachelor of Computer Engineering

Work Experience

CareCloud Jul 2025 – Present
AI Engineer

- Automated workflows for 3+ healthcare RCM applications, reducing manual effort by 60% and improving operational efficiency.
- Developed AI-driven healthcare solutions using Python and LLM technologies, improving processing accuracy and user experience.
- Designed scalable backend architectures using asynchronous programming, multiprocessing, and multithreading, reducing processing time by 40%.

Automotive Artificial Intelligence (AAI) Feb 2025 – Jun 2025
AI Developer

- Engineered an AI-powered text auto-completion system using Large Language Models (LLMs) and FastAPI, improving text generation efficiency by 70% and reducing manual input time.
- Built and optimized backend services for user profile management, authentication, and session handling using FastAPI, supporting scalable API performance and secure user access.

CARE Pvt. Ltd. – Center for Advanced Research in Engineering Aug 2024 – Jan 2025
Software Intern

- Designed and optimized Seller and Customer Portal workflows, improving UX and streamlining end-to-end platform operations.
- Architected a scalable database architecture by designing an ERD with 30+ entities covering users, products, orders, and transactions.
- Trained and deployed ML/DL models (GANs, ResNet, MLP, Random Forest) for gemstone classification and image generation, achieving up to 88% accuracy with data preprocessing and augmentation.

Projects

Synthetic Image Generation and Identification of Gemstones Oct 2023 – May 2024
Final Year Project

- Built Vanilla, Conditional, and DCGAN models to generate synthetic gemstone images, achieving an FID score of 46.
- Trained a CNN for gemstone classification, reaching 88% accuracy on real and generated images.
- Prototyped IoT hardware with Bluetooth control to capture high-quality gemstone images for model training.

Financial News Analysis and Stock Prediction Jun 2024 – Aug 2024

- Analyzed correlation between financial news sentiment and stock market trends using NLP on 10,000+ financial news articles.
- Performed EDA and implemented an LSTM-based stock prediction model leveraging sentiment-driven financial features.

AI Virtual Assistant Chatbot: Jessi Nov 2024 – Dec 2024

- Created a voice-enabled AI assistant using the Gemini API, supporting 10+ voice commands for automated user interactions.
- Integrated web navigation, real-time news retrieval, and music playback, consolidating 3+ productivity features into a single assistant.

Traceability Solution for Hierarchical Relationships Nov 2024 – Feb 2025

- Developed a FastAPI-based backend for multi-level hierarchical traceability, support mapping across 1,000+ interconnected records.
- Containerized and deployed the solution using Docker, Kubernetes, and MongoDB, enabling scalable and reliable data persistence.

Technical Skills

Languages: Python, C++, SQL, MIPS, Dart/Flutter, LaTeX

Technologies: FastAPI, GANs, LLMs, Kubernetes, Flask, Data Structures & Algorithms

Tools: MongoDB, Kaggle, Docker, GCP, MATLAB, MySQL, JIRA, Arduino IDE, Git/GitHub, OpenTelemetry

Libraries: TensorFlow, Keras, PyTorch, LangChain, Scikit-learn, Pandas

Certifications

- Certificate of **Machine Learning Specialization** — DeepLearning.AI & Stanford via Coursera
- Kaggle Certification in **Pandas Library**.
- KPMG — Data Analytics Consulting Virtual Internship.
- Docker for Developers — LinkedIn Learning.

Publication

- Development and Industrial Implementation of Pakistan's First Integrated Biofuel Production Process: A Pathway Toward Green Energy Transition**, *The China International UGS Academic Conference*, October 2025.